The Witness Vessel: A Machine-Checked Lossless Transport Protocol on the Dual-Link SIC-POVM

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Abstract

A paraconsistent kernel pair, the Lean 4 library `p4ramill` and the bare-metal operating system `mOMonadOS`, carries structural verdicts of formally typed conjectures (the "Witnesses") between composition universes. Earlier transport programs lost information, and the loss is reproducible: the tensor (cargo) reading of transport is structurally denied, raising a cross-primitive axiom violation at the Ð ↔ Þ pair. This paper presents the corrected protocol, in which a Witness rides *as* the vessel's own split/fuse structure rather than inside it, and verifies its losslessness twice, on two substrates that share a design but share no toolchain and extend no trust to one another. On the proof substrate, the round-trip law $\mu \circ \delta = \mathrm{id}$ is a kernel-checked theorem whose proof rests on `propext` alone, and the payload verdicts are computed by `native_decide` from canonical data rather than asserted: Birch–Swinnerton-Dyer boards as T, Hodge as T, and Yang–Mills as the dialetheia B, the value that splits into (T, F) and rides both arms. The bundled theorem `witness_vessel_lossless` additionally delivers the vessel intact at the destination: the transported fiducial still satisfies both defining SIC-POVM conditions, riding the recently completed $d = 12$ existence theorem without re-deriving it. On the runtime substrate, the same protocol executes on bare metal in QEMU over a payload of three Witnesses evaluated under all 88 composition universes; every boarding action closes under the Frobenius verification harness, the verdict matrix is identical before and after transport, and the run reports $\Delta S = 0$. Losslessness here means literal equality after read-back; nothing physical is claimed. The theorem is generic over Belnap payload vectors, of which the Clay verdicts are one instantiation.

Contents

*Thanks are owed to Harry T. Larson; cf. [2].

1 Introduction

Two artifacts implement the same twelve-token algebra. The Lean 4 library `p4ramill` (inside the `p4rakernel` repository) states and machine-checks its theorems; the operating system `mOMonadOS`, a freestanding Rust kernel for x86-64, executes the same structure as its native control flow: Belnap four-valued registers, fork/join execution frames for the split and fuse tokens, and a verification harness that enforces $\mu(\delta(q)) = q$ after every action before the execution loop advances. There is no Lean 4 toolchain inside the operating system and nothing in the operating system is trusted by the Lean 4 kernel. When the two agree, neither is taking the other's word for it.

Both artifacts type conjectures as twelve-primitive imscriptions and evaluate them under *composition universes*: rulesets, each consisting of three ordinal gates, a temporal constitution, and absorption rules. A *Witness* is the structural verdict a typed conjecture earns under a universe. Earlier *witness-drag* programs moved these verdicts between universes, and the transports were lossy. The present work identifies the loss mechanism, replaces the transport, and proves and then observes that the replacement loses nothing.

The two headline results:

1. **Proof half** (section 4). In the Lean 4 module `SIC_D12_WitnessVessel` the boarding map is $\delta = $ `fsplit` and the read-back map is $\mu = $ `ffuse`, and the law

$$\forall\, p, \quad \mu(\delta(p)) = p$$

holds for *every* list of Belnap values. Its proof depends on the axiom `propext` alone. The concrete payloads are computed, not chosen: the gate verdicts of the three Clay Witnesses are derived by `native_decide` from their canonical catalog tuples and the frozen universe rulesets. Yang–Mills derives to B, the dialetheia: its gates close while its temporal ceiling blocks, and $\texttt{fsplit}(\mathsf{B}) = (\mathsf{T}, \mathsf{F})$, so this Witness literally rides both arms of the vessel. The bundle `witness_vessel_lossless` adds self-reference preservation and delivery: conjugation commutes with transport on all twelve vessel coordinates, and the transported $d = 12$ fiducial still satisfies unit norm and equiangularity at the destination.

2. **Runtime half** (section 6). The module `witness_vessel.rs` runs the same protocol on bare metal. The payload is deliberately wide: each of the three Witnesses is evaluated under all 88 composition universes, each universe judging by its own gates and its own temporal constitution, giving a 3×88 verdict matrix. Every entry boards through two independent execution substrates (the belief-register virtual machine and the kernel's fork frames), every boarding action is gated by the Frobenius harness, and the matrix is recomputed after read-back. In QEMU the run closes every check, reports $\Delta S = 0$, and finds the matrices identical.

Section 7 states precisely what is and is not claimed. The short version: "lossless" means equality after read-back, "universe" is a technical term for a composition ruleset, and "transport" is a pair of maps. The interest of the result lies in where the equality comes from (a Frobenius law available at the lowest axiomatic cost the proof assistant offers) and in the two-substrate agreement, not in any physical reading.

2 Preliminaries

2.1 Belnap FOUR and the split/fuse pair

Belnap's four-valued logic [1] has carriers $\{\mathsf{N}, \mathsf{T}, \mathsf{F}, \mathsf{B}\}$: no information, true only, false only, and both. The knowledge order places N at the bottom and B at the top, with join written $\vee_k$. The kernel pair uses one distinguished map pair on this lattice:

$$\texttt{fsplit}(s) = \begin{cases} (\mathsf{T}, \mathsf{F}) & s = \mathsf{B} \\ (s, s) & \text{otherwise} \end{cases} \qquad \texttt{ffuse}(a, b) = a \vee_k b.$$

Lemma 2.1 (`split_fuse_id`)**.** *For every s, $\texttt{ffuse}(\texttt{fsplit}(s)) = s$.*

The only nontrivial case is the dialetheia: $\mathsf{B} \mapsto (\mathsf{T}, \mathsf{F}) \mapsto \mathsf{T} \vee_k \mathsf{F} = \mathsf{B}$. The converse also pins the decomposition: $\texttt{fsplit}(s) = (\mathsf{T}, \mathsf{F})$ forces $s = \mathsf{B}$ (`only_B_splits_to_TF`). These are existing theorems of the frozen module `BelnapSplitFuse`; the vessel adds no new axioms to use them.

2.2 Imscriptions, universes, Witnesses

A structural type is imscribed as a twelve-primitive tuple over the primitives Đ, Þ, Ř, Φ, ƒ, Ç, Γ, ɢ, ⊙, Ħ, Σ, Ω, each valued in a small family of Shavian values with canonical ordinals. The three Witnesses of this paper are the canonical catalog entries

Birch–Swinnerton-Dyer	⟨ıʃɾƕϱɩɿʃɔƺϱϛ⟩
Hodge	⟨ıϐɒƕɛɩɿʃɔJɀϛ⟩
Yang–Mills	⟨ɭʃɾƸʅɿɿʃɣJɀɜ⟩

sourced procedurally from the canonical catalog (`IG_catalog.json`); no tuple in this paper was written by hand.

A *composition universe* is a ruleset: three gates, each requiring one primitive to sit at or above an ordinal threshold; a temporal constitution ("T-side"), a set of per-primitive critical values checked either exactly or as ceilings; and absorption rules, which play no role here. The runtime carries 88 universes: 8 canonical, then hand-crafted and expansion families. Two temporal conventions matter below:

- **T_CANONICAL** (per-universe default): exact match on four dynamic primitives and a ceiling on Ç only, anchored at Φ = ꝺ, ʃ = ʅ, Ç ≤ ɩ, Ħ = v, Ω = ϛ.
- **T_CEILING** (the Clay convention): all five as ceilings at the same anchors. This is the convention of the frozen module `Clay_WitnessedClosure` and of the runtime's Clay gates.

A Witness's verdict under a universe is a single Belnap value combining the gate layer and the T-side:

Definition 2.2 (`layerVerdict`). *For booleans g (gates closed) and t (T-side consistent),*

$$\mathtt{layerVerdict}(g,t) = \begin{cases} \mathsf{T} & g \wedge t \\ \mathsf{B} & g \wedge \neg t \\ \mathsf{N} & \neg g \wedge t \\ \mathsf{F} & \neg g \wedge \neg t. \end{cases}$$

The B row is the point: a Witness whose gates close while its T-side blocks is not "roughly true" or "roughly false"; it is exactly the dialetheia, and the vessel must carry that value without flattening it.

2.3 The frozen substrate

The vessel re-derives nothing. It imports three frozen components:

- `SIC_D12_Embedding`: the $d = 12$ SIC-POVM existence capstone. Its theorem `crystal_forces_d12_sic` is axiom-audit-clean (the standard trio `propext`, `Classical.choice`, `Quot.sound`, plus `Lean.ofReduceBool` and `Lean.trustCompiler` for `native_decide`); the star ring homomorphism $\varphi : R \to \mathbb{C}$ from the existence ring $R = K_{16}(s_0, s_1, s_3, s_9, i, c_5, u_1)$ transfers unit norm and all 143 equiangularity identities. The companion manuscript in this repository records that construction; here it is furniture.
- `Clay_WitnessedClosure`: the structural closure results for BSD and Hodge (five closer universes each, T_CEILING-consistent) and the Yang–Mills "one bump short" verdict (gate-closed under triple-criticality, T_CEILING-blocked on Ç alone), all by `native_decide` over canonical tuples.
- `BelnapSplitFuse`: the split/fuse pair and its laws.

3 The loss mechanism, and the design provenance

This section reports procedural outputs of the Grammar tooling. They are design rationale for the protocol, recorded for provenance; they are not offered as mathematical proof, which sections 4 and 6 carry alone.

Why the drags were lossy. The natural first reading of transport is cargo: put the Witness in a container, tensor it with the vessel. That reading is denied, and the denial is univocal. The tensor of the vessel type with a Witness type is undefined (six primitive conflicts), and every cofactor factorization attempt raises a cross-primitive axiom violation at the Ð ↔ Þ pair: the request itself is malformed, and the Grammar localizes the malformation. A transport implemented against the cargo reading loses information because there is no well-formed object for it to be.

The mirror reading. The corrected reading pairs the vessel with the B-state directly: the mirror distance evaluates to 4.5 with asymmetry within numerical zero of the metric's floor, and the proof path between the pair has length two *in both directions*, with the same two operations that close the two closed Clay Witnesses: apply modularity; restrict to Hodge degree $(1, 1)$. A round trip that is write-in and read-back with the same moves is the signature of a lossless carrier.

Independent reconstruction. A six-entity design batch for the protocol was generated and validated end to end: six of six entities valid, six of six Frobenius-closed, with entropy change reported as $\Delta S \approx 0$ throughout. The designer, given only the transport brief, independently produced the boarding decomposition (Witness → [modulus, phase] → Witness) as the Frobenius content of three separate entities, and typed the historical lossy path and the corrected path as the F and T arms of a single dialetheia-complete protocol. The split/fuse boarding of section 4 is therefore not a gloss written onto the design after the fact; it is the design's own structural core, reconstructed independently of the Lean development.

4 The transport theorem (proof half)

All results in this section are theorems of `SIC_D12_WitnessVessel`, compiled green in the full library build.

4.1 Payloads are derived, not asserted

The three Witness verdicts are *computed* from the frozen closure results via definition 2.2:

```
def bsdVerdict   : Belnap := layerVerdict bsdGateClosed   (tCeilingConsistent bsd)
def hodgeVerdict : Belnap := layerVerdict hodgeGateClosed (tCeilingConsistent hodge)
def ymVerdict    : Belnap := layerVerdict ymGateClosed    (tCeilingConsistent ym)

theorem bsdVerdict_T   : bsdVerdict   = .T := by native_decide
theorem hodgeVerdict_T : hodgeVerdict = .T := by native_decide
theorem ymVerdict_B    : ymVerdict    = .B := by native_decide
```

Listing 1: Payload derivation (excerpt).

Here `bsdGateClosed` conjoins the operad-layer closure of BSD under its five closer universes (chirality-first, scope, kinetics-trap, and the two absorption variants), and likewise for Hodge; `ymGateClosed` is closure under triple-criticality. The third theorem is the one to notice: nobody wrote B anywhere. Yang–Mills closes at the gate layer and fails T_CEILING on Ç alone (its kinetics value ı exceeds the ι anchor, visible directly in its tuple in section 2), so its verdict *derives* to the dialetheia.

4.2 The round-trip law

Boarding maps a payload pointwise through `fsplit`; read-back maps through `ffuse`.

Theorem 4.1 (`roundtrip`)**.** *For every list p of Belnap values,* $\mathit{readback}(\mathit{board}(p)) = p$.

Theorem 4.2 (`b_cargo_mechanism`)**.** $\mathit{fsplit}(\mathit{ymVerdict}) = (T, F)$ *and* $\mathit{ffuse}(T, F) = \mathit{ymVerdict}$.

Theorem 4.1 is proved by list induction on top of `split_fuse_id`, and its axiom audit returns `[propext]`: the losslessness law of the vessel costs one axiom, the cheapest the proof assistant sells. The concrete payload instance (`mpp_roundtrip`) is separately checked by `native_decide`.

4.3 Self-reference travels inside

The vessel's own coordinates carry their conjugation structure through transport. With φ the frozen star ring homomorphism of the $d = 12$ capstone and $\bar{z}_k$ the ring-side conjugate of coordinate z_k:

Theorem 4.3 (`coords_selfref`)**.** *For all* $k < 12$*:* $\varphi(\bar{z}_k) = \overline{\varphi(z_k)}$.

This is the vessel principle in one line: conjugation is *internal* to the carried data (the cover phases are real by construction, so the star commutes with the hom), which is exactly what failed in the cargo reading.

4.4 The bundle

Theorem 4.4 (`witness_vessel_lossless`)**.** *All of the following hold: (i)* $\mu \circ \delta = \mathrm{id}$ *on every Belnap payload; (ii)* $\mathit{ymVerdict} = B$*; (iii) the concrete MPP payload round-trips; (iv) self-reference is preserved on all twelve coordinates; (v) the transported fiducial has unit norm; and (vi) it satisfies equiangularity,* $(d + 1)\, \|\langle \psi, D_{a,b}\, \psi \rangle\|^2 = 1$ *for all* 143 *non-identity displacements at* $d = 12$.

Conjuncts (v) and (vi) are the delivery condition: the vessel does not merely return its cargo, it *arrives*, still satisfying both defining SIC-POVM conditions at the destination, by direct reuse of the frozen capstone theorems. The audit of the bundle is the standard trio plus the `native_decide` pair; no project axioms, no Stark shadow, no circularity.

5 The boarding mechanism is the Dual-Link structure

The protocol's split/fuse pair is not an arbitrary choice of bijection; it is the same structure the $d = 12$ construction identified as the Dual-Link SIC-POVM, read at the Belnap layer.

- **B-collapse is the modulus.** In the existence ring, the pairing of a coordinate with its conjugate collapses to a real modulus: $z_k \bar{z}_k = N_k$ in the totally real base field. The dialetheia behaves identically at the logic layer: B splits into the pair (T, F) whose fuse is again B; the pair is the "both arms" carry, and the join is the collapse.
- **The cover phase keeps conjugation internal.** The capstone builds its cover generators as real square roots, so conjugation never leaves the carried data; `coords_selfref` is that fact restated as a transport invariant.
- **Half-angle read-back.** The phase u_1 is reconstructed from carried real data by the half-angle identities; read-back of the phase from the modulus-side data is a theorem, not a measurement.
- **Hom-independence is universe-independence.** The ring identities hold in R; *any* star homomorphism $R \to \mathbb{C}$ transports them (the flip audit of the existence campaign made this exact: every branch choice lands on a genuine SIC point). The destination supplies the hom; the cargo does not depend on which.

This correspondence is why the paper speaks of one vessel rather than two protocols that happen to share a diagram.

6 The runtime half: bare metal, all universes

6.1 Substrate

`mOMonadOS` is a freestanding Rust kernel. Three of its components matter here, all of which predate the vessel:

- `kernel.rs`: graph execution over the twelve tokens, with `FSPLIT` pushing a fork frame (the un-gated right arm carries a value across the fork) and `FFUSE` joining the arms by Belnap join.
- `parasm.rs`: a belief-register virtual machine whose `FSPLIT` instruction bifurcates B to (T, F) on its two destination registers, the exact semantics of the Lean 4 `fsplit`.
- `frob_verify.rs`: the Frobenius harness. Every action δ must verify $\mu(\delta(q)) = q$ before the loop advances. The operating system was built around the invariant the vessel needs.

6.2 The wide payload: three Witnesses under all 88 universes

The runtime evaluates each Witness under *all* 88 composition universes, each universe judging by its own three gates and its own temporal constitution, through definition 2.2. Ordinal comparisons use the canonical ordinal table (never raw enum discriminants, which are non-monotonic for three families). Alongside the wide matrix, the runtime computes the proof half's mirror trio exactly as section 4 defines it: closer-universe gates plus T_CEILING.

Every one of the resulting values boards through *both* execution substrates, each boarding action is checked by the harness, and the full matrix is recomputed from the canonical tuples after read-back.

6.3 Observed run

Booted in QEMU, the `vessel run` command reports (excerpt):

```
-- Payloads (computed from canonical tuples + closer universes) --
  BSD    verdict T (Lean: T)  transport T -> T   [OK]
  Hodge  verdict T (Lean: T)  transport T -> T   [OK]
  YM     verdict B (Lean: B)  transport B -> B   [OK]

-- Wide payload: 3 witnesses x 88 universes --
  BSD    N:1  T:0  F:75 B:12
  Hodge  N:0  T:2  F:74 B:12
  YM     N:2  T:0  F:85 B:1

-- Round trip (both substrates, frob_verify gated) --
  Boarding actions checked: 534 (ParaVM + ForkFrame per value)
  Frobenius closed: 534  open: 0  closure ratio: 1.0
  Delta-S (total mismatches): 0
  Gate verdicts before == after: YES (all entries)

VERDICT: LOSSLESS -- mu-after-delta = id on every payload.
```

Listing 2: QEMU serial output (excerpt).

Three features of the matrix deserve note.

1. **The Yang–Mills row holds exactly one B, at triple-criticality.** Of 88 universes, precisely the one whose gates are three criticality rungs closes Yang–Mills at the gate layer while its T-side blocks. The dialetheia is not spread thinly across the expansion; it is localized at the single universe the Clay analysis had already identified.
2. **BSD reads B at exactly its five closer universes.** Under the wide matrix's per-universe convention (T_CANONICAL, exact match on four primitives), BSD's fidelity value ϙ fails the exact ℩ anchor, so where its gates close the verdict is B; under the Clay convention (T_CEILING, all ceilings) the same structure reads T. Two honest readings of one structure, both transported without loss. The protocol is indifferent to which convention the analyst prefers; it carries either.
3. **Nothing was tuned.** The 88 universes were defined independently of the vessel (the expansion landed as data before this protocol existed); the vessel is simply their first consumer.

6.4 What the runtime does and does not check

The runtime checks are executions, not proofs: they demonstrate that on this boot, with these tuples, every action closed and the matrices agreed. The generality (*every* payload, not just the observed ones) lives only in theorem 4.1. Conversely, the proof half cannot observe that a physical machine's fork frames actually behave as modeled; only the run shows that. The two halves are complementary in exactly the way the two-substrate architecture intends.

7 Honest scope

For the curmudgeon, plainly:

- **"Lossless" means equality.** The claim is $\mu(\delta(p)) = p$, literally, plus before/after equality of recomputed verdict matrices. The design layer's "$\Delta S \approx 0$" appears in the theorems as =, nothing weaker.

- **Nothing physical is claimed.** "Universe" is a composition ruleset over a finite type lattice; "transport" is a pair of maps; "boarding" is function application. No claim about spacetime, quantum mechanics, or actual travel of any kind is made or implied.
- **The Clay verdicts are structural, not mathematical.** That BSD earns T under these rulesets is a fact about the Grammar's operad model of the conjecture's type, machine-checked as such. It is not a proof of BSD, and this paper does not move any Millennium problem.
- **The payloads are one instantiation.** Theorem 4.1 quantifies over all Belnap payload vectors. The Clay Witnesses give the protocol its motivation and its test data, nothing more; a reader who rejects the conjecture-typing program entirely may read this paper as a verified-transport result over four-valued verdict vectors and lose only the motivation.
- **The provenance section is not evidence.** The Grammar verdicts and the design batch of section 3 explain where the protocol came from. The theorems and the run stand without them.
- **What would falsify the claim.** Any open result from the Frobenius harness, any before/after matrix mismatch, any nonzero ΔS, or any axiom in the audit beyond the declared five. None occurred.

8 Discussion

The lossy drags failed because they tried to put the Witness *in* the vessel, and the Grammar's refusal of that reading was univocal: the tensor is undefined and the factorization request is malformed at exactly the Ð ↔ Þ pair. The corrected protocol needs no container because the Witness's verdict already has the vessel's shape: the dialetheia is the value that decomposes into a (T, F) pair and reconstitutes under join, which is the Belnap-layer image of the Dual-Link pairing $z\bar{z} = N$ with its ramified double cover. Ride *as*, not *in*.

Epistemically, the result is a small instance of a pattern this project uses deliberately: prove the law on the substrate built for proving, run the law on the substrate built for running, and let agreement carry the weight. The proof half's `[propext]` audit says the transport law is nearly definitional; the runtime half says a machine with no access to that proof lands in the same place, five hundred and thirty-four gated actions without one opening.

Two directions follow naturally. First, wider payload families: the 3×88 matrix is the first object computed across the full universe expansion, and its structure (the localization of B values, the convention-dependence of T readings) invites systematic study. Second, the abstraction path: since the theorem is generic over payloads, the protocol applies unchanged to any verdict-valued analysis that wants a transport with a machine-checked loss bound of zero.

9 Artifact statement

The proof half is the module `SIC_D12_WitnessVessel.lean` in the `p4ramill` library of the `p4rakernel` repository, building green in the full-library `lake build`; its axiom audits are reproducible with `#print axioms`. The runtime half is `src/witness_vessel.rs` in the `mOMonadOS` repository, exercised via the `vessel run` serial command under QEMU. The canonical tuples come from `IG_catalog.json`; the 88 universes from `universe_expansion.rs`. No tuple, verdict, or payload in either half is hand-entered.

10 Acknowledgements

Thanks are owed to Harry T. Larson [2].

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